EV40-F

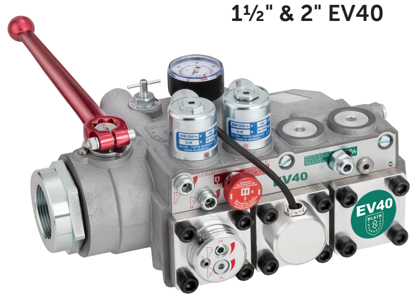
VVVF Driven Elevator Control Valves

EN ISO 9001

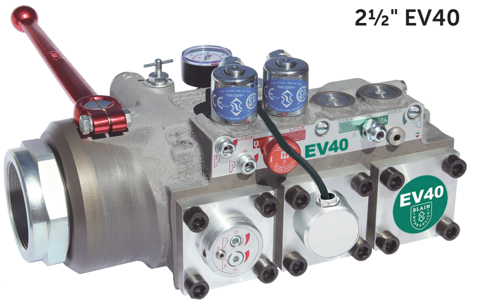
The BLAIN EV40-vvvf program includes the widest range of vvvf solution offered to the elevator industry for high performance passenger elevators. Easy to install, EV40's are smooth, reliable and precise in operation throughout extreme load and temperature variations with inbuilt overload protection and different energy saving modes. The EV40 system uses the L1000H or GA700 VVVF drive to control up travel, while down travel is managed by the EV40 valve itself. In this way, the EV40-vvvf solution offers a most cost-effective and energy-efficient alternative.

¾" EV40

1½" & 2" EV40



2½" EV40



Description

Available port sizes are ¾", 1½", 2", and 2½" pipe threads, depending on flow requirements. The **EV40-F** eliminates high inrush currents and does not require wye-delta switching. Based on customer elevator data, valves are factory-adjusted, ready for operation, and very easy to readjust if needed. The **L1000H** or **GA700 YASKAWA** drives, combined with feedback systems, are designed to compensate for elevator speed fluctuation, regardless of oil temperature or car load conditions.

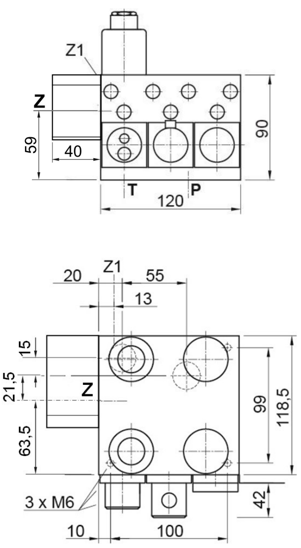
Caution: The EV4 valve is to be used only together with L1000H or GA700 YASKAWA drives and not as a standalone control valve. EV40 valves include the following features essential for efficient installation and trouble free service:

B44.1
C US
ASME-A17.1

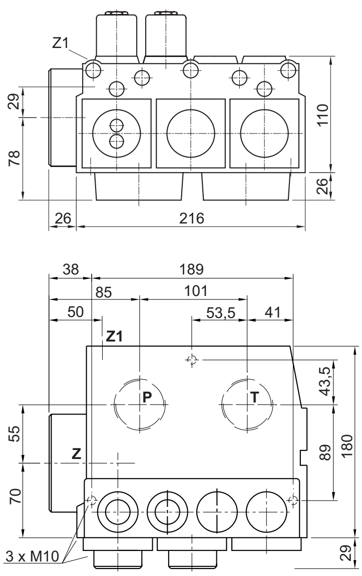
- Simple Responsive Adjustment
- Temperature and Pressure Compensations
- Pressure Gauge and Shut Off Cock
- Self Closing Manual Lowering
- Self Cleaning Pilot Line Filters
- Self Cleaning Main Line Filter (Z-T)
- Built-in Turbulence Suppressors
- 70 HRc Rockwell Hardened Bore Surfaces
- 100% Continuous Duty Solenoids
- Compact and aesthetic design

Technical Data:		¾" EV40	1½" & 2" EV40	2½" EV40
Flow Range:	l/min (US gpm)	10-125 (2-33)	30-800 (8-212)	500-1530 (130-405)
Pressure Range (valve):	bar (psi)	8-70 (116-1015)	8-70 (116-1015)	8-68 (116-986)
Press. Range CSA (valve):	bar (psi)	8-55 (117-797)	8-55 (117-797)	8-55 (117-797)
Burst Pressure Z:	bar (psi)	575 (8340)	505 (7324)	340 (4931)
Pressure Drop P–Z:	bar (psi)	6 (87) at 125 l/min	4 (58) at 800 l/min	4 (58) at 1530 l/min
Weight:	kg (lbs)	5 (11)	10 (22)	14 (31)
Coils AC:	24 V/1.8 A, 42 V/1.0 A, 110 V/0.43 A, 230 V/0.18 A, 50/60 Hz.			
Coils DC:	12 V/2.0 A, 24 V/1.1 A, 42 V/0.5 A, 48 V/0.6 A, 80 V/0.3 A, 110 V/0.25 A, 196 V/0.14 A.			
Oil Viscosity:	25-75 cSt. at 40°C (104°F).			
Operation oil temperature range:	10°C-60°C (50°F-140°F), for oil VGA46: 250cSt.-20 cSt.			
Optimal oil temperature range:	20°C-55°C (68°F-131°F), for oil VGA46: 140cSt.-24 cSt.			
Ambient temperature range:	0°C-50°C (32°F-122°F)			
Insulation Class, AC and DC:	IP 68			
Max. Oil Temperature:	70°C (158°F)			

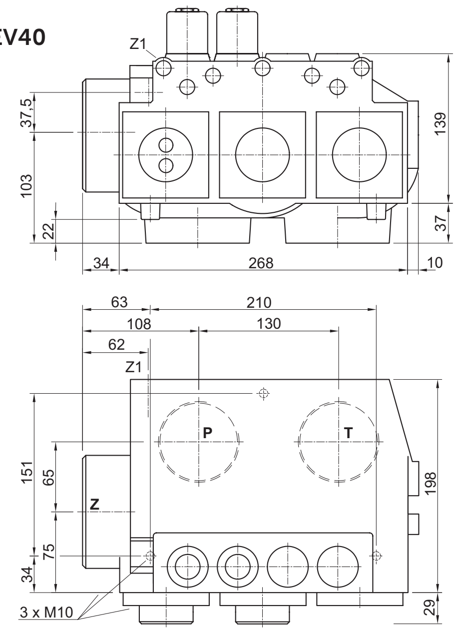
¾" EV40



1½" & 2" EV40



2½" EV40



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BLAIN HYDRAULICS GmbH

Designer and Manufacturer of the highest quality control valves & safety components for hydraulic elevators

Optional Equipment

EN	Emergency Power Coil	**DH**	High Pressure Switch
CSA	CSA Coils	**DL**	Low Pressure Switch
KS	Slack Rope Valve	**CX**	Pressure Compensated Down Valve
BV	Main Shut-Off Valve	**MX**	Auxiliary Down
HP	Hand Pump	**iL10-S**	UCM valve

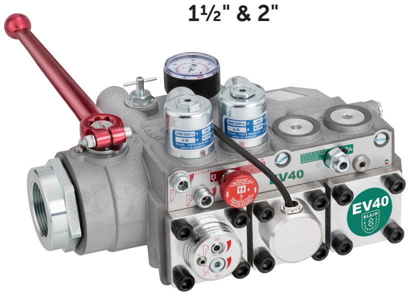
C SP US B44.1 ASME-A17.1

CE

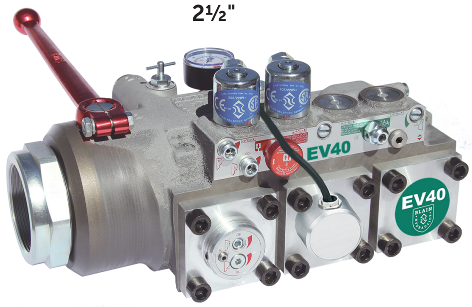
EV40

¾"

1½" & 2"



2½"



Up Up to 1m/s (200 fpm). 1 full speed and 1 levelling speed. All up-direction input data and setup are done via a mobile phone.

Down Up to 1 m/s (200 fpm). 1 Full Speed and 1 Levelling Speed. All down functions are smooth and adjustable.

Control Elements

C Solenoid (Down Deceleration)
D Solenoid (Down Stop)
H Manual Lowering
S Relief Valve

U By Pass Valve
V Check Valve
X Full Speed Valve (Down)
Y Levelling Valve (Down)
F Filter

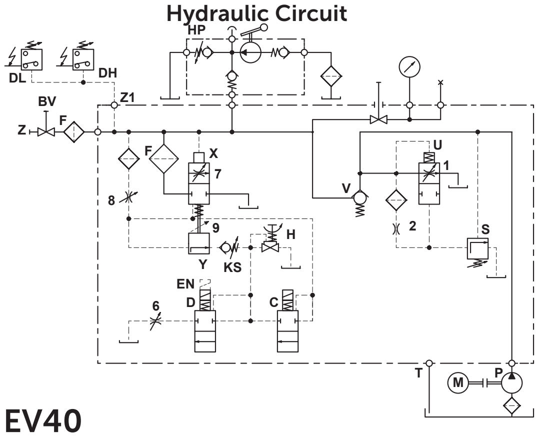
Adjustments UP

None
(Fixed Orifice)

Adjustments DOWN

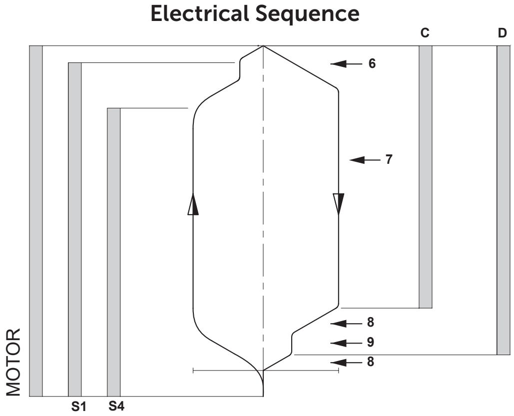
6 Down Acceleration
7 Down Full Speed
8 Down Deceleration
9 Down Levelling Speed

Hydraulic Circuit



EV40

Electrical Sequence

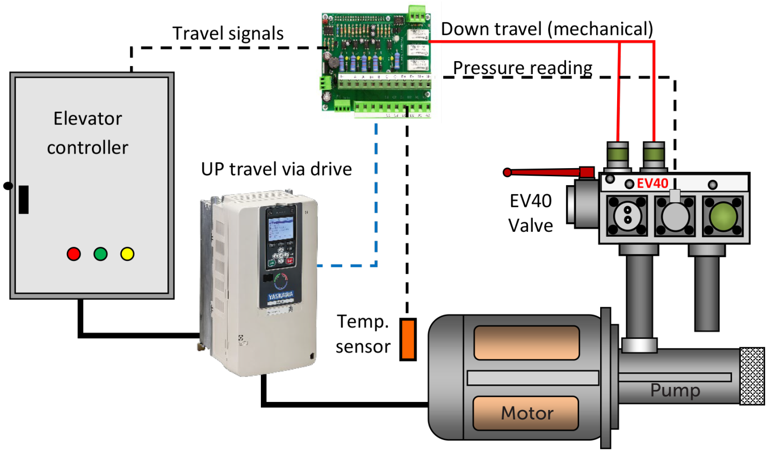


UP direction control

Caution: Refer to the detailed installation and set-up procedures in the EV40-F Handbook and the YASKAWA L1000H or GA700 Technical Manual.

The upward direction is controlled by the YASKAWA L1000H or GA700 drives. Using its software, the EV40-F system reads the car's load and current oil temperature, then processes the data to determine the appropriate motor speeds for nominal, inspection, and levelling travel.

The system can be set up quickly and easily via a smartphone, without needing to adjust any drive parameters. An advanced teach run with an empty car is sufficient for self-configuration, ensuring smooth take-off and accurate stopping at the floor.



Warning: Only qualified personnel should adjust or service the EV40 valve and the L1000H or GA700 drives. Unauthorised manipulation may result in injury, loss of life or damage to equipment. Prior to servicing internal parts, ensure that the electrical controller is switched off, cylinder line is closed and residual pressure in the valve is reduced to zero.

DOWN travel adjustments

Valves are already adjusted and tested. Check electrical operation before changing valve settings. Test that the correct coil is energised, by removing nut and raising the coil slightly to feel pull.

Standard settings: Adjustments 7 and 9 should be set level with the flange face. Then, turn adjustment 9 out by half a turn. Turn adjustments 6 and 8 fully inward. For EV40 ¾": Turn adjustment 6 out by 2½ turns and adjustment 8 out by 1 turn. For EV40 1½"–2½": Turn adjustment 6 out by 2 to 2½ turns and adjustment 8 out by 1½ turns.

6. Down Acceleration: When coils **C** and **D** are energized, the car will accelerate downwards according to the setting of adjustment **6**. 'In' (clockwise) provides a softer down acceleration, 'out' (c-clockwise) a quicker acceleration.

7. Down Speed: With coils **C** and **D** energized as in **6** above, the full down speed of the car is according to the setting of adjustment **7**. 'In' (clockwise) provides a slower down speed, 'out' (c-clockwise) a faster down speed.

8. Down Deceleration: When coil **C** is de-energized whilst coil **D** remains energized, the car will decelerate according to the setting of adjustment **8**. 'In' (clockwise) provides a softer deceleration, 'out' (c-clockwise) a quicker deceleration.
Attention: Do not close all the way in! Closing adjustment 8 completely (clockwise) may cause the car to fall on the buffers.

9. Down Levelling: With coil **C** de-energized and coil **D** energized as in **8** above, the car will proceed at its down levelling speed according to the setting of adjustment **9**. 'In' (clockwise) provides a slower, 'out' (c-clockwise) a faster down levelling speed.

Down Stop: When coil **D** is de-energized with coil **C** remaining de-energized, the car will stop according to the setting of adjustment **8** and no further adjustment will be required.

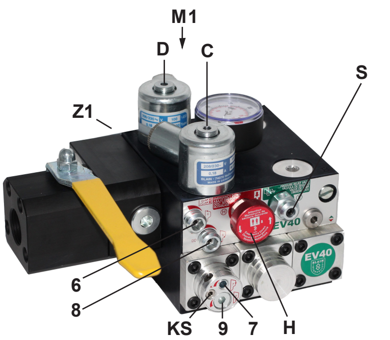
KS Slack Rope Valve: Both coils **C** and **D** must be de-energized beforehand! Loosen the small grub screw located on the left-hand side of the down flange. The **KS** is adjusted using a 3 mm Allen key. Turn **in** for higher pressure and **out** for lower pressure. With KS turned fully in, then backed out half a turn, the unloaded car should descend when Manual Lowering **H** is opened. If the car does not descend, turn **KS** further out slowly until the car just begins to descend. Then turn it out an additional half turn to ensure the car can be lowered properly, even with cold oil.

Adjustments pressure relief valve

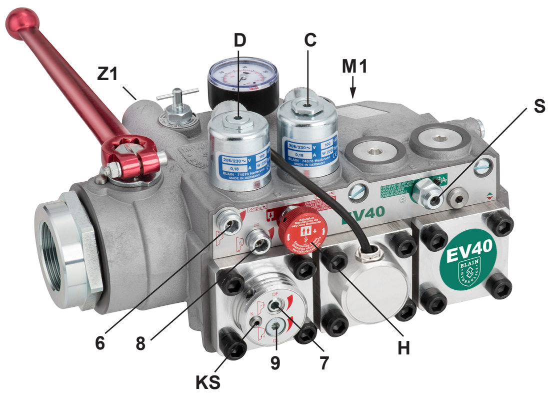
Valves are pre-adjusted for proper functionality. Verify electrical operation before making any changes to the inverter settings. Refer to the EV40 manual for the required parameter settings related to upward travel.

S Relief Valve: Turning **'in'** (cw) increases and turning **'out'** (ccw) decreases the **maximum pressure setting**. After turning the valve **out**, briefly open Manual Lowering **H** to read the actual pressure.

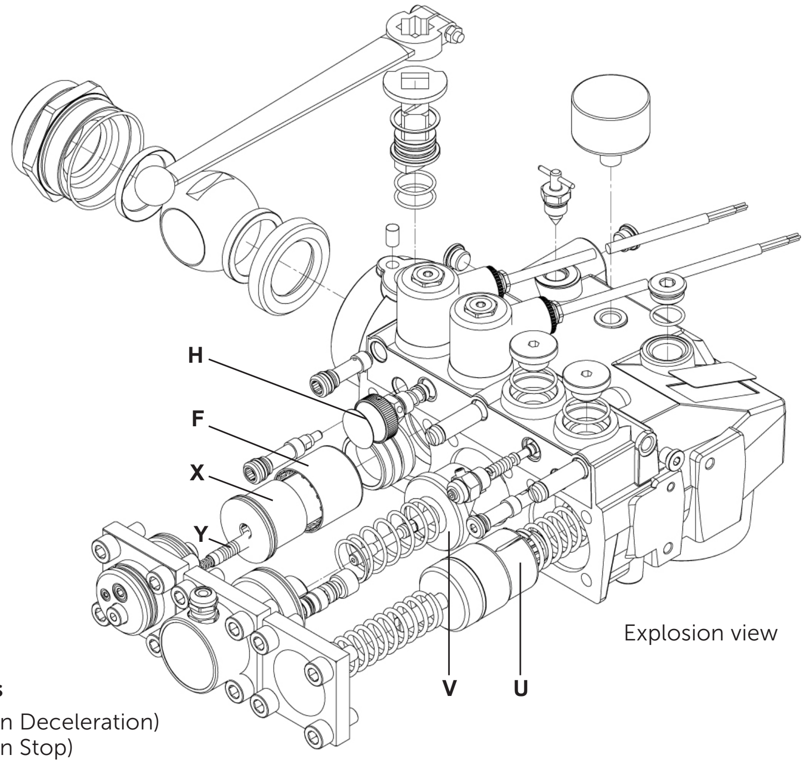
Important: When testing the relief valve, always close the ball valve gradually to avoid sudden pressure spikes.



M1 Second pressure gauge connection, ½"
Z1 Pressure switch connection, ¼"



Important: Length of ¾" thread on pump connector should not be longer than 14 mm!



Explosion view

Adjustments DOWN

6 Down Acceleration
7 Down Full Speed
8 Down Deceleration
9 Down Levelling Speed

Control Elements

C Solenoid (Down Deceleration)
D Solenoid (Down Stop)
H Manual Lowering
S Relief Valve
U By Pass Valve
V Check Valve
X Full Speed Valve (Down)
Y Levelling Valve (Down)
2 Fix Orifice

EV40 Spare Parts List

Pos.	No.	Item
1	FS	Lock Screw - Flange
	FO	O-Ring - Flange
	1F4	Flange - By Pass
	UO	O-Ring - By Pass Valve
	U4	By Pass Valve
	UD	Noise Suppressor
	UF1	Spring - By Pass
	UF2	Spring - By Pass
	US	Dead Stop
2	2	Fixed orifice
3	3	Plug
4	4F4	Flange - Check Valve
	FO	O-Ring - Flange
	VF	Spring - Check Valve
	VO	Seal - Check Valve
	V	Check Valve
	W	Up-Levelling Valve
	WO	O-Ring - Up Levelling Valve
	VO	Seal - Check Valve
	W6	Screw - Check Valve
5	3	Plug
6	3	Adjustment - Down Acceleration
7	7F	Flange - Down Valve
	FO	O-Ring - Flange
	7O	O-Ring - Adjustment
	7E	Adjustment - Down Valve
	UO	O-Ring - Down Valve
	XO	Seal - Down Valve
	X	Down Valve
	XD	Noise Suppressor
	F	Main Filter
8	8	Adjustment - Down Deceleration
9	EO	O-Ring - Adjustment
	9E	Adjustment - Down Levelling
	9F	Spring - Down Valve
	Y	Down Levelling Valve
H	H	Manual Lowering - Self Closing
	HO	Seal - Manual Lowering
S	SE	Adjustment - Screw
	SM	Hexagonal
	MS	Grub Screw
	SO	O-Ring - Nipple
	SZ	Nipple
	SF	Spring
	SK	Piston
C+D	MM	Nut - Solenoid
	M	Coil - Solenoid (indicate voltage)
	DR	Tube - Solenoid 'Down'
	MO	O-Ring - Solenoid
	DF	Spring - Solenoid 'Down'
	DN	Needle - 'Down'
	DK	Core - Solenoid
	DG	Seat Housing with Screen-'Down'
	FD	Filter Solenoid
	DS	Seat - Solenoid 'Down'

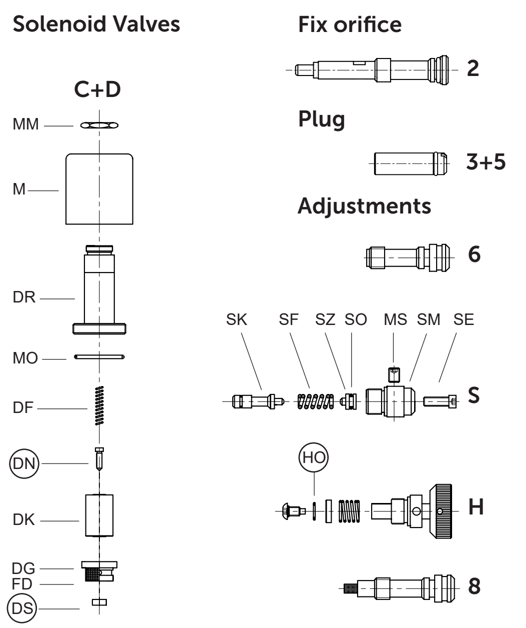
Some parts occur more than once in different positions of the valve.

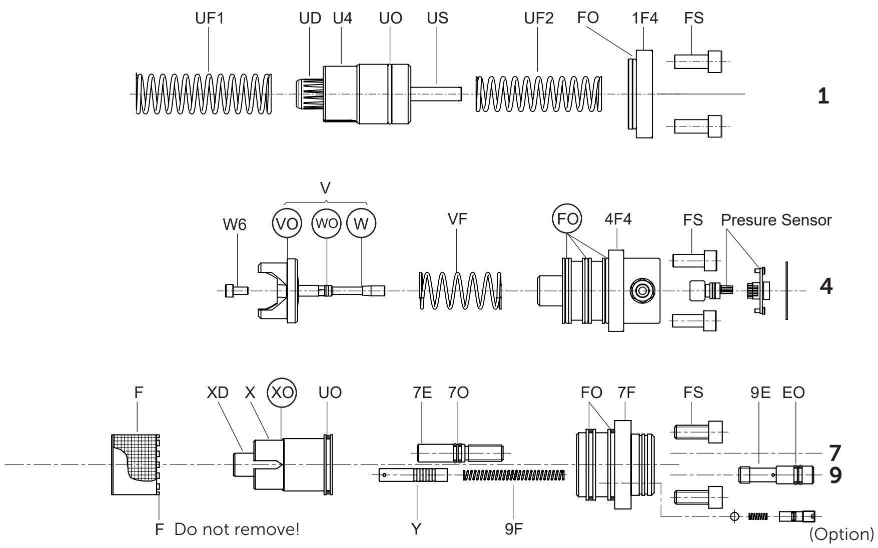
O-Ring-Size			
No.	¾"	1½"	2½"
FO	26x2P	47x2.5P	58x3P *
EO	9x2P	9x2P	9x2P
UO	26x2V	39.34x2.62V	58x3V
WO	5.28x1.78V	5.28x1.78V	5.28x1.78V
VO	23x2,5V	42x3V	60x3V **
7O	5.28x1.78P	9x2P	9x2P
XO	13x2V	30x3V	47x3V
HO	5.28x1.78V	5.28x1.78V	5.28x1.78V
SO	5.28x1.78P	5.28x1.78P	5.28x1.78P
MO	26x2P	26x2P	26x2P

* FO by 4F 2 ½" is 67x2.5P
** 90 Shore

O-Ring: V = FKM - Viton
P = NBR - Perbunan

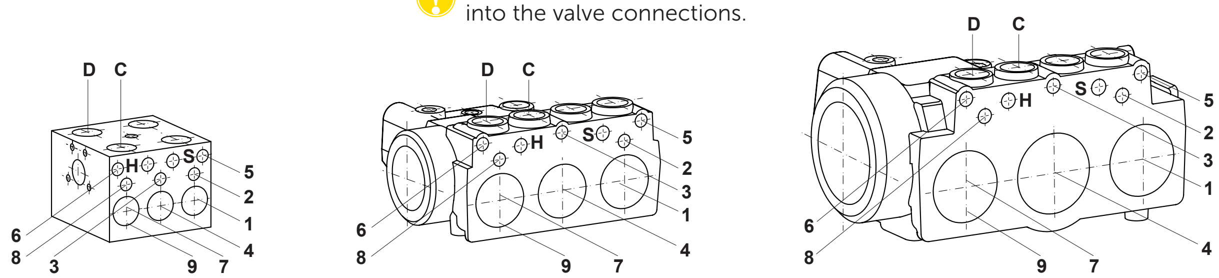
US is only for EV40 1½" and above sizes!



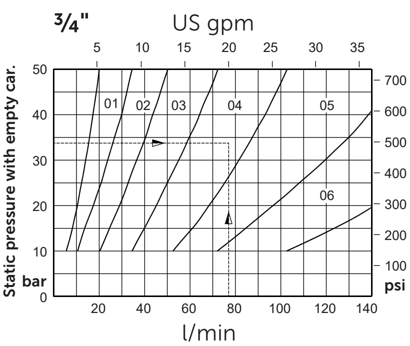


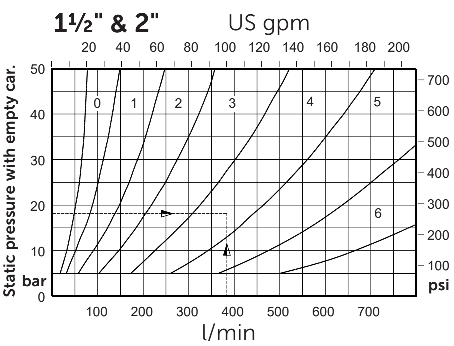
In case of internal leakage, replace and test in the following order: DS & DN, XO, VO, WO, FO + HO.

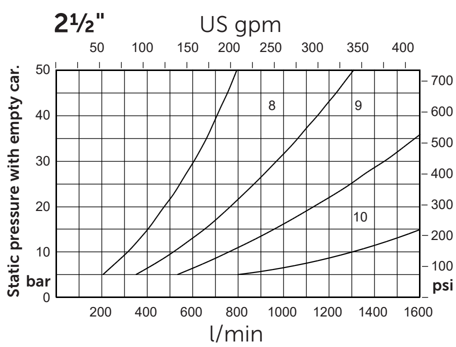
Taper threads: Do not exceed 8 turns of piping into the valve connections.



Flow Guide Selection Charts







To order EV40: Size (inch), state pump flow, empty car pressure (or flow guide size) and coil voltage.
Example order: 1½"EV40, 380l/min, 18bar (empty), 110AC or 1½"EV40/4/110AC